

ECE456 Network Centric Programming – TCP Concepts

1. (8 points) A client executes:

```
1 send(sock, "A", 1, 0);  
2 send(sock, "B", 1, 0);  
3 send(sock, "C", 1, 0);
```

The server executes:

```
1 recv(sock, buf, 3, 0);
```

Which results are possible?

- ABC
- A
- AB
- AC
- BCA

2. (8 points) TCP provides:

- message boundaries
- byte stream

3. (10 points) A program sends:

```
1 send(sock, buf, 1000, 0);
```

Which are possible values returned?

- 1000
- 500
- 1
- -1

Circle all that apply.

4. (10 points) The client sends:

```
1 send(sock, "HELLO", 5, 0);
```

The server receives:

```
1 recv(sock, buf, 100, 0);
```

What values can `recv()` return?

- 5
- 100
- 3
- -1

5. (8 points) TCP guarantees:

- ordered delivery
- reliable delivery
- fixed packet sizes
- constant latency

Circle all correct answers.

6. (10 points) A TCP ACK indicates:

- the packet arrived at the remote TCP stack
- the server application processed the message
- the data reached the remote network card
- the message was written to disk

Circle the correct answer.

7. (8 points) A client sends 100 bytes.

The server calls:

```
1 recv(sock, buf, 50, 0);
2 recv(sock, buf, 50, 0);
```

Is it guaranteed that both calls return 50 bytes?

- Yes
- No

8. (10 points) A server executes:

```
1 int n = recv(sock, buf, 1000, 0);
```

Which return value means the connection has closed?

- -1
- 0
- 1
- 1000

9. (12 points) Consider:

```
1 char buf[100];
2
3 int n = recv(sock, buf, 100, 0);
4
5 buf[n] = '\0';
6
7 printf("%s\n", buf);
```

What bug occurs if `n == 100`?

(single phrase)

10. (12 points) Which of the following can cause `recv()` to return fewer bytes than requested?

- network packet boundaries
- TCP buffering
- sender used multiple `send()` calls
- receiver buffer too small

Circle all that apply.

11. (15 points) A client sends three messages:

```
1 send(sock, "AAA", 3, 0);  
2 send(sock, "BBB", 3, 0);  
3 send(sock, "CCC", 3, 0);
```

The server calls:

```
1 recv(sock, buf, 9, 0);
```

List two possible values of the received data.